LAB SHEET – 02

BCA201P – Foundations of Java Development
BCA III Semester | Session 2026–2027

Complete Answers: Questions 1–30

Each question includes a Java program and a sample output. Run each program separately using a Java IDE or save the code as Main.java. For the random-number program, output will vary on each run.

Note: These are typed solutions for reference. Check your instructor's requirements and test each program before submitting.

Q1. Area of a triangle using Heron’s formula

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three sides: ");
        double a = sc.nextDouble(), b = sc.nextDouble(), c = sc.nextDouble();
        if (a + b <= c || a + c <= b || b + c <= a) {
            System.out.println("Invalid triangle");
        } else {
            double s = (a + b + c) / 2.0;
            double area = Math.sqrt(s * (s-a) * (s-b) * (s-c));
            System.out.printf("Area = %.2f%n", area);
        }
        sc.close();
    }
}
```

Sample Output

```
Enter three sides: 3 4 5
Area = 6.00
```

Q2. Compound interest

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter principal, annual rate (%) and years: ");
        double p = sc.nextDouble(), r = sc.nextDouble(), t = sc.nextDouble();
        double amount = p * Math.pow(1 + r / 100.0, t);
        System.out.printf("Amount = %.2f%n", amount);
        System.out.printf("Compound Interest = %.2f%n", amount - p);
        sc.close();
    }
}
```

Sample Output

```
Enter principal, annual rate (%) and years: 10000 5 2
Amount = 11025.00
Compound Interest = 1025.00
```

Q3. Distance between two points

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter x1, y1, x2, y2: ");
        double x1=sc.nextDouble(), y1=sc.nextDouble();
        double x2=sc.nextDouble(), y2=sc.nextDouble();
        double d = Math.sqrt(Math.pow(x2-x1, 2) + Math.pow(y2-y1, 2));
        System.out.printf("Distance = %.2f%n", d);
        sc.close();
    }
}
```

Sample Output

```
Enter x1, y1, x2, y2: 1 2 4 6
Distance = 5.00
```

Q4. Prefix and postfix increment/decrement visitor counter

Source Code

```
public class Main {
    public static void main(String[] args) {
        int visitors = 0;
        System.out.println("Initial visitors: " + visitors);
        System.out.println("Entering (prefix ++): " + (++visitors));
        System.out.println("Entering (postfix ++): " + (visitors++));
        System.out.println("After entry: " + visitors);
        System.out.println("Leaving (prefix --): " + (--visitors));
        System.out.println("Leaving (postfix --): " + (visitors--));
        System.out.println("After leaving: " + visitors);
    }
}
```

Sample Output

```
Initial visitors: 0
Entering (prefix ++): 1
Entering (postfix ++): 1
After entry: 2
Leaving (prefix --): 1
Leaving (postfix --): 1
After leaving: 0
```

Q5. Negative value using unary operator

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n = sc.nextInt();
        System.out.println("Negative value = " + (-n));
        sc.close();
    }
}
```

Sample Output

```
Enter an integer: 8
Negative value = -8
```

Q6. Repeatedly halve a number using /=

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a positive number: ");
        double n = sc.nextDouble();
        int steps = 0;
        while (n >= 1) {
            n /= 2;
            steps++;
        }
        System.out.println("Steps = " + steps);
        System.out.println("Final value = " + n);
        sc.close();
    }
}
```

Sample Output

```
Enter a positive number: 20
Steps = 5
Final value = 0.625
```

Q7. Total rainfall for seven days using +=

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double total = 0;
        for (int day = 1; day <= 7; day++) {
            System.out.print("Rainfall on day " + day + ": ");
            total += sc.nextDouble();
        }
        System.out.println("Total rainfall = " + total);
        sc.close();
    }
}
```

Sample Output

```
Rainfall on day 1: 2
Rainfall on day 2: 0
Rainfall on day 3: 4
Rainfall on day 4: 1
Rainfall on day 5: 3
Rainfall on day 6: 0
Rainfall on day 7: 2
Total rainfall = 12.0
```

Q8. Check whether three angles form a valid triangle

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three angles: ");
        int a=sc.nextInt(), b=sc.nextInt(), c=sc.nextInt();
        if (a > 0 && b > 0 && c > 0 && a+b+c == 180)
            System.out.println("Valid triangle");
        else
            System.out.println("Invalid triangle");
        sc.close();
    }
}
```

Sample Output

```
Enter three angles: 60 60 60
Valid triangle
```

Q9. Lexicographic comparison of strings using character relations

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two strings: ");
        String a=sc.next(), b=sc.next();
        int i=0, result=0;
        while (i<a.length() && i<b.length()) {
            char x=a.charAt(i), y=b.charAt(i);
            if (x < y) { result=-1; break; }
            if (x > y) { result=1; break; }
            i++;
        }
        if (result==0) {
            if (a.length() < b.length()) result=-1;
            else if (a.length() > b.length()) result=1;
        }
        if (result < 0) System.out.println(a + " comes before " + b);
        else if (result > 0) System.out.println(a + " comes after " + b);
        else System.out.println("Both strings are equal");
        sc.close();
    }
}
```

Sample Output

```
Enter two strings: apple banana
apple comes before banana
```

Q10. Check leap year and whether it is in a given range

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter year, range start and range end: ");
        int year=sc.nextInt(), start=sc.nextInt(), end=sc.nextInt();
        boolean leap = (year%400==0) || (year%4==0 && year%100!=0);
        if (leap && year>=start && year<=end)
            System.out.println("Leap year within range");
        else
            System.out.println("Condition not satisfied");
        sc.close();
    }
}
```

Sample Output

```
Enter year, range start and range end: 2024 2000 2030
Leap year within range
```

Q11. Course pass using logical operators

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter theory %, practical % and overall %: ");
        double theory=sc.nextDouble(), practical=sc.nextDouble(), overall=sc.nextDouble();
        if ((theory>=40 && practical>=50) || overall>=50)
            System.out.println("Pass");
        else
            System.out.println("Fail");
        sc.close();
    }
}
```

Sample Output

```
Enter theory %, practical % and overall %: 42 52 47
Pass
```

Q12. Smallest of four numbers using nested ternary operators

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter four numbers: ");
        int a=sc.nextInt(), b=sc.nextInt(), c=sc.nextInt(), d=sc.nextInt();
        int ab = (a < b) ? a : b;
        int cd = (c < d) ? c : d;
        int smallest = (ab < cd) ? ab : cd;
        System.out.println("Smallest = " + smallest);
        sc.close();
    }
}
```

Sample Output

```
Enter four numbers: 8 3 5 1
Smallest = 1
```

Q13. Classify a character using the ternary operator

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch=sc.next().charAt(0);
        String type = Character.isDigit(ch) ? "Digit" :
            ((ch>='a' && ch<='z') || (ch>='A' && ch<='Z') ?
            ("aeiouAEIOU".indexOf(ch)>=0 ? "Vowel" : "Consonant") :
            "Special symbol");
        System.out.println(type);
        sc.close();
    }
}
```

Sample Output

```
Enter a character: e
Vowel
```

Q14. Swap two integers using XOR

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two integers: ");
        int a=sc.nextInt(), b=sc.nextInt();
        a = a ^ b;
        b = a ^ b;
        a = a ^ b;
        System.out.println("After swap: a = " + a + ", b = " + b);
        sc.close();
    }
}
```

Sample Output

```
Enter two integers: 5 9
After swap: a = 9, b = 5
```

Q15. Count set bits using bitwise operators

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a non-negative integer: ");
        int n=sc.nextInt(), count=0;
        while (n>0) {
            count += n & 1;
            n >>>= 1;
        }
        System.out.println("Number of set bits = " + count);
        sc.close();
    }
}
```

Sample Output

```
Enter a non-negative integer: 13
Number of set bits = 3
```

Q16. Multiply and divide by powers of two using shifts

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer and shift count: ");
        int n=sc.nextInt(), k=sc.nextInt();
        System.out.println("Multiplication by 2^k = " + (n << k));
        System.out.println("Division by 2^k = " + (n >> k));
        sc.close();
    }
}
```

Sample Output

```
Enter an integer and shift count: 8 2
Multiplication by 2^k = 32
Division by 2^k = 2
```

Q17. Rotate a 32-bit integer left by two positions

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n=sc.nextInt();
        int rotated = (n << 2) | (n >>> 30);
        System.out.println("After left rotation by 2 bits = " + rotated);
        sc.close();
    }
}
```

Sample Output

```
Enter an integer: 1
After left rotation by 2 bits = 4
```

Q18. First 20 Fibonacci terms using for loop

Source Code

```
public class Main {
    public static void main(String[] args) {
        int a=0, b=1;
        for (int i=1; i<=20; i++) {
            System.out.print(a + (i<20 ? " " : "\n"));
            int next=a+b;
            a=b;
            b=next;
        }
    }
}
```

Sample Output

```
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181
```

Q19. Armstrong numbers between 1 and 1000

Source Code

```
public class Main {
    public static void main(String[] args) {
        for (int n=1; n<=1000; n++) {
            int temp=n, digits=0, sum=0;
            do { digits++; temp/=10; } while (temp>0);
            temp=n;
            while (temp>0) {
                int digit=temp%10;
                sum += (int)Math.pow(digit, digits);
                temp/=10;
            }
            if (sum==n) System.out.print(n + " ");
        }
    }
}
```

Sample Output

```
1 2 3 4 5 6 7 8 9 153 370 371 407
```

Q20. Accept password until correct using do-while

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        final String correctPassword = "java123";
        String entered;
        do {
            System.out.print("Enter password: ");
            entered=sc.nextLine();
        } while (!entered.equals(correctPassword));
        System.out.println("Access granted");
        sc.close();
    }
}
```

Sample Output

```
Enter password: hello
Enter password: java123
Access granted
```

Q21. Display all factors using do-while

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a positive integer: ");
        int n=sc.nextInt(), i=1;
        if (n>0) {
            do {
                if (n%i==0) System.out.print(i + " ");
                i++;
            } while (i<=n);
        } else {
            System.out.println("Please enter a positive integer.");
        }
        sc.close();
    }
}
```

Sample Output

```
Enter a positive integer: 12
1 2 3 4 6 12
```

Q22. Reverse digits using while loop

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n=sc.nextInt(), reverse=0;
        int temp=Math.abs(n);
        while (temp>0) {
            reverse=reverse*10 + temp%10;
            temp/=10;
        }
        if (n<0) reverse=-reverse;
        System.out.println("Reversed number = " + reverse);
        sc.close();
    }
}
```

Sample Output

```
Enter an integer: 1234
Reversed number = 4321
```

Q23. Check palindrome number using while loop

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a non-negative integer: ");
        int n=sc.nextInt(), temp=n, reverse=0;
        while (temp>0) {
            reverse=reverse*10 + temp%10;
            temp/=10;
        }
        if (n>=0 && n==reverse) System.out.println("Palindrome");
        else System.out.println("Not a palindrome");
        sc.close();
    }
}
```

Sample Output

```
Enter a non-negative integer: 121
Palindrome
```

Q24. Maximum and minimum in an array using for-each

Source Code

```
public class Main {
    public static void main(String[] args) {
        int[] values = {12, 5, 27, -3, 18};
        int max=values[0], min=values[0];
        for (int value : values) {
            if (value>max) max=value;
            if (value<min) min=value;
        }
        System.out.println("Maximum = " + max);
        System.out.println("Minimum = " + min);
    }
}
```

Sample Output

```
Maximum = 27
Minimum = -3
```

Q25. Average marks stored in a 2D array using for-each

Source Code

```
public class Main {
    public static void main(String[] args) {
        int[][] marks = {{80, 75, 90}, {70, 85, 65}, {88, 92, 78}};
        int total=0, count=0;
        for (int[] student : marks) {
            for (int mark : student) {
                total += mark;
                count++;
            }
        }
        System.out.printf("Average marks = %.2f%n", (double)total/count);
    }
}
```

Sample Output

```
Average marks = 80.33
```

Q26. Weekday or weekend using switch-case

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter day number (1=Monday, 7=Sunday): ");
        int day=sc.nextInt();
        switch (day) {
            case 1: case 2: case 3: case 4: case 5:
                System.out.println("Weekday"); break;
            case 6: case 7:
                System.out.println("Weekend"); break;
            default:
                System.out.println("Invalid day number");
        }
        sc.close();
    }
}
```

Sample Output

```
Enter day number (1=Monday, 7=Sunday): 7
Weekend
```

Q27. Meaning of grade using switch-case

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter grade (A-F): ");
        char grade=Character.toUpperCase(sc.next().charAt(0));
        switch (grade) {
            case 'A': System.out.println("Excellent"); break;
            case 'B': System.out.println("Very Good"); break;
            case 'C': System.out.println("Good"); break;
            case 'D': System.out.println("Average"); break;
            case 'E': System.out.println("Pass"); break;
            case 'F': System.out.println("Fail"); break;
            default: System.out.println("Invalid grade");
        }
        sc.close();
    }
}
```

Sample Output

```
Enter grade (A-F): A
Excellent
```

Q28. Print 1 to 50, skipping perfect squares using continue

Source Code

```
public class Main {
    public static void main(String[] args) {
        for (int n=1; n<=50; n++) {
            int root=(int)Math.sqrt(n);
            if (root*root==n) continue;
            System.out.print(n + " ");
        }
    }
}
```

Sample Output

```
2 3 5 6 7 8 10 11 12 13 14 15 17 18 19 20 21 22 23 24 26 27 28 29 30 31 33 34 35 36 37 38 39 40 41 42 43 45 46 47 48 49
```

Q29. Generate random numbers until divisible by 7 and 13

Source Code

```
import java.util.Random;
public class Main {
    public static void main(String[] args) {
        Random random = new Random();
        int n;
        do {
            n=random.nextInt(100)+1;
            System.out.println("Generated: " + n);
            if (n%7==0 && n%13==0) break;
        } while (true);
        System.out.println("Found: " + n);
    }
}
```

Sample Output

```
Sample output (random values vary):
Generated: 17
Generated: 42
Generated: 91
Found: 91
```

Q30. Mixed advanced: power of 4, toggle 3rd bit, table with continue and break

Source Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a positive integer: ");
        int n=sc.nextInt();

        int temp=n;
        boolean powerOfFour = temp>0;
        while (temp>1 && powerOfFour) {
            if ((temp & 3) != 0) powerOfFour=false;
            temp >>= 2;
        }
        System.out.println("Power of 4: " + powerOfFour);

        int toggled = n ^ (1 << 2); // Toggle third bit (bit position 2)
        System.out.println("After toggling 3rd bit: " + toggled);

        for (int i=1; i<=20; i++) {
            int product=n*i;
            if (product%48==0) {
                System.out.println("Reached multiple of 48: " + product);
                break;
            }
            if (product%6==0) continue;
            System.out.println(n + " x " + i + " = " + product);
        }
        sc.close();
    }
}
```

Sample Output

```
Enter a positive integer: 4
Power of 4: true
After toggling 3rd bit: 0
4 x 1 = 4
4 x 2 = 8
Reached multiple of 48: 48
```